

WEEK 3 – CYBERSECURITY PRACTICAL REPORT

PASSWORD CRACKING USING JOHN THE RIPPER AND NETWORKWALKS TOOLS

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Program: Cybersecurity Internship Program – Networkwalks

Week: 03

Practical Modules: Project Module 1 and Project Module 2

Operating Environment: Windows PC

1. LIABILITY DISCLAIMER

The practical activities described in this report were carried out strictly for educational and cybersecurity training purposes using the file provided as part of the authorized Networkwalks laboratory exercise.

Password-cracking techniques can be used for legitimate security assessments, password-strength testing, and cybersecurity education. However, attempting to recover passwords from files, accounts, or systems without proper authorization may be unlawful.

The tools and techniques demonstrated in this report must therefore be used only within an approved scope and on systems or files for which permission has been granted.

No unauthorized system, account, or personal document was targeted during these activities.

2. INTRODUCTION

Password protection is an important security mechanism used to prevent unauthorized access to confidential files. However, the effectiveness of password protection depends greatly on the strength of the password being used.

As part of the Week 3 cybersecurity practical, I performed two password-security exercises using the same authorized password-protected PDF file, **My Locked PDF1.pdf**.

The first exercise involved **John the Ripper (JTR)** and its graphical interface, **Johnny**. The PDF password information was extracted and provided to JTR for password recovery.

The second exercise used the **Networkwalks Hash Calculator** and **Networkwalks Password Cracker**. In this approach, the PDF was processed through the online hash calculator and the resulting hash was supplied to the password-cracking tool.

Both approaches successfully recovered the password used to protect the laboratory PDF. The recovered password was then entered into the original file to verify the result.

The practical provided an understanding of the relationship between password-protected files, password hashes, and password-recovery techniques.

3. OBJECTIVES

The major objectives of the practical were:

1. To understand the fundamentals of password cracking.
2. To understand how password-protected PDF files can be analyzed.
3. To learn the purpose of password hashes.
4. To gain practical experience with John the Ripper.
5. To understand the use of Johnny as a graphical interface for JTR.
6. To extract a PDF password hash.
7. To use the Networkwalks Hash Calculator.
8. To use the Networkwalks Password Cracker.
9. To verify a recovered password against the original protected PDF.
10. To understand the security risks associated with weak passwords.
- 11.

4. CONCEPTUAL BACKGROUND

4.1 Password Cracking

Password cracking refers to the process of attempting to recover a password from password-related information stored in a protected system or file.

In cybersecurity, password-cracking techniques are commonly used in authorized environments to evaluate whether passwords provide adequate protection.

The purpose of such testing is not simply to recover passwords, but to identify weaknesses in password choices and improve overall security.

4.2 Password Hash

A hash is a processed representation of information that can be used during password verification.

In the case of the practical PDF, the protected file was processed to obtain information representing the password protection. The extracted value was then supplied to a password-cracking tool.

The extracted PDF hash used during the practical began with the \$pdf\$ identifier.

4.3 John the Ripper

John the Ripper is a password-security auditing and password-recovery tool widely used in cybersecurity laboratories.

It supports multiple password formats and can be used to test the strength of passwords associated with different types of protected data.

For this practical, JTR was used to process the extracted PDF password information.

4.4 Johnny

Johnny provides a graphical interface for John the Ripper.

Instead of performing every operation directly through the command line, users can interact with JTR through graphical options.

This makes the password-testing workflow easier to understand for users who are new to John the Ripper.

5. PROJECT MODULE 1 – PASSWORD CRACKING WITH JTR

5.1 Practical Description

The first practical focused on recovering the password of the authorized PDF file using John the Ripper and Johnny.

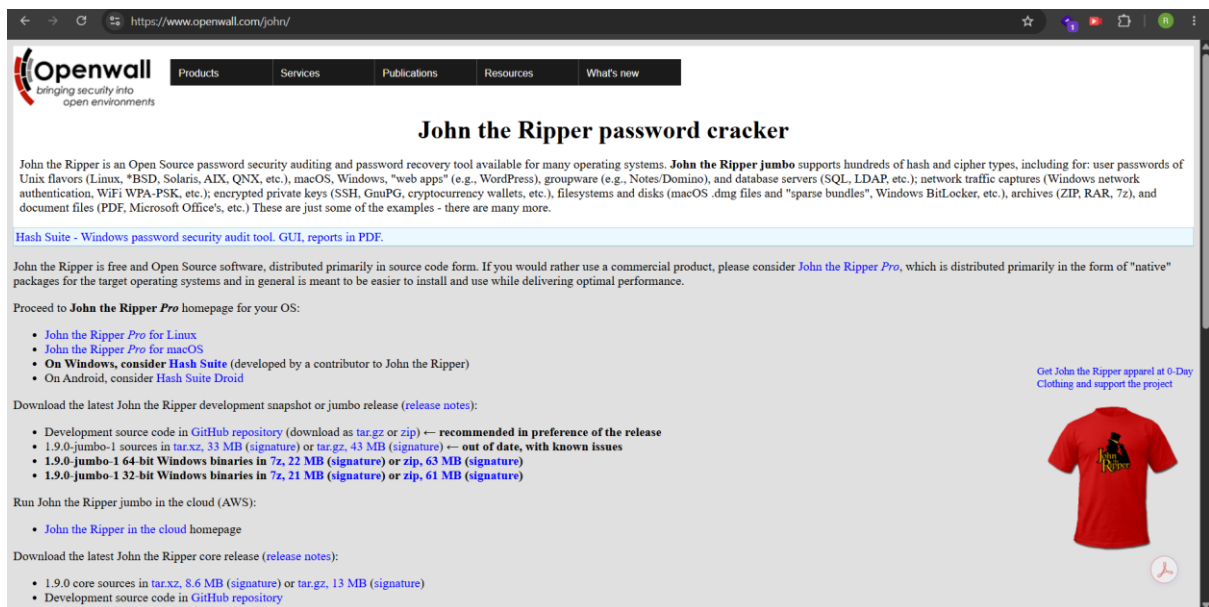
The workflow consisted of obtaining the PDF hash, storing the hash in a text file, loading the file into Johnny, starting the password-recovery process, and finally verifying the recovered password.

5.2 JTR Installation

John the Ripper was obtained for the Windows environment from the Openwall project.

After obtaining the required JTR package, the relevant files were made available on the Windows system.

The installation/setup process was completed before starting the password-recovery activity.



	01. JTR_John CLI Win x64	Date modified: 28-08-2026 23:27
	01. JTR_John CLI Win x64.zip	Date modified: 28-08-2026 15:03

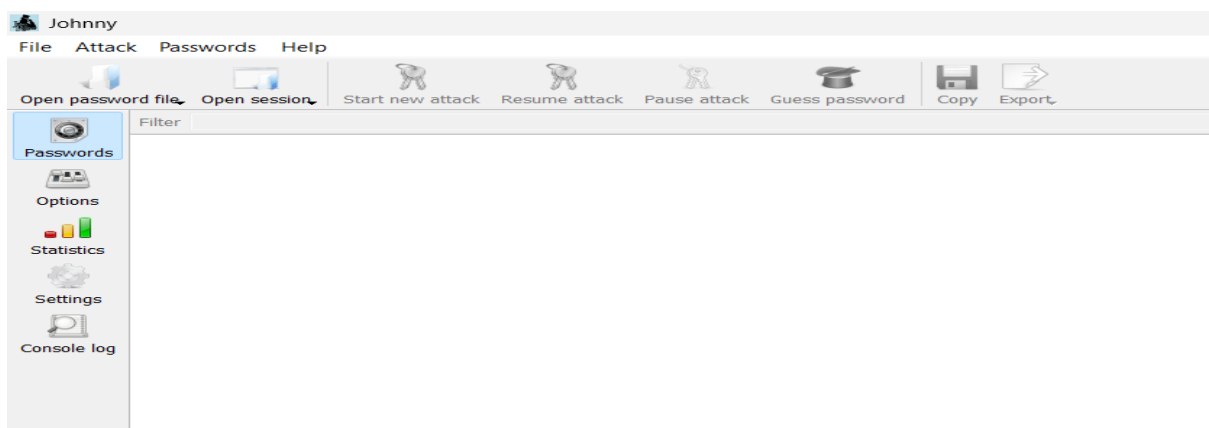
5.3 Johnny GUI Setup

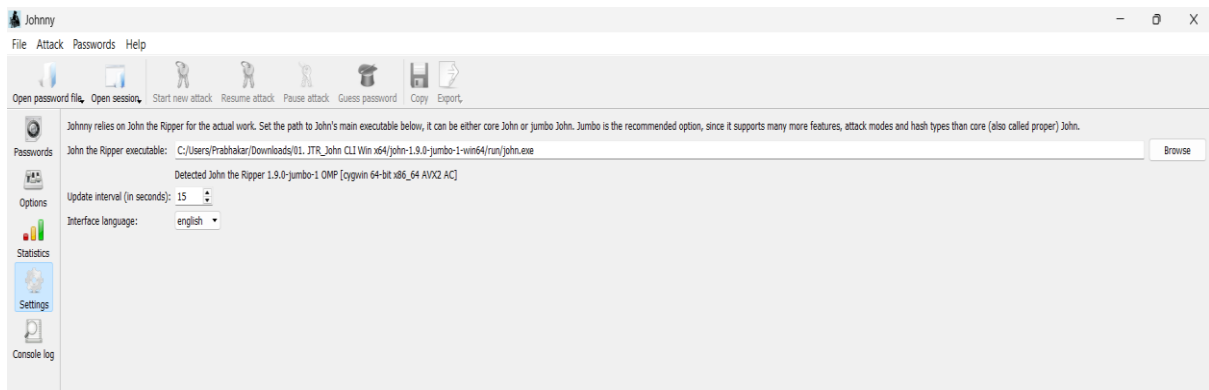
Johnny was installed and opened on the Windows system.

The settings section of Johnny was used to specify the location of the John the Ripper executable.

The john.exe file located inside the JTR run directory was selected.

This configuration allowed Johnny to communicate with the John the Ripper engine





 johnny.exe	27-08-2026 22:20	Application	886 KB
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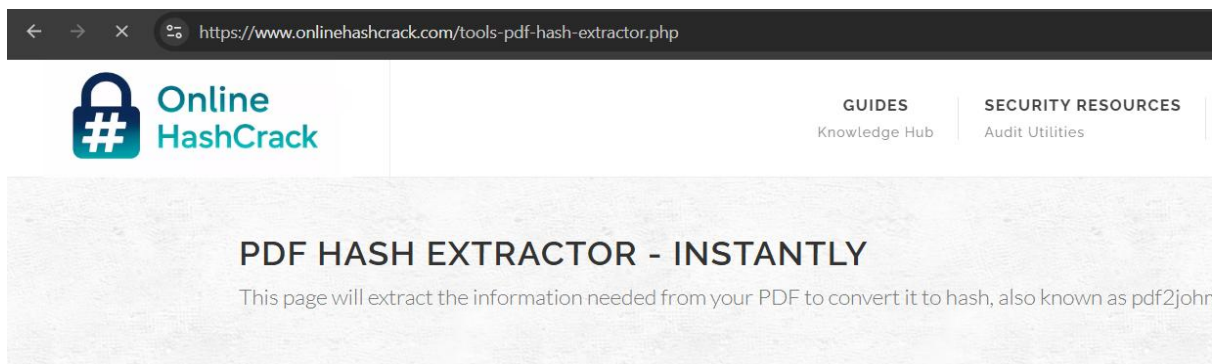
5.4 Preparing the PDF Hash

The provided My Locked PDF1.pdf file was processed using the assigned PDF hash extraction website.

The file was uploaded through the hash extraction interface.

After processing, a PDF password hash was generated.

The complete hash was copied for use in the JTR workflow.



Select and Upload your PDF:

Select file...

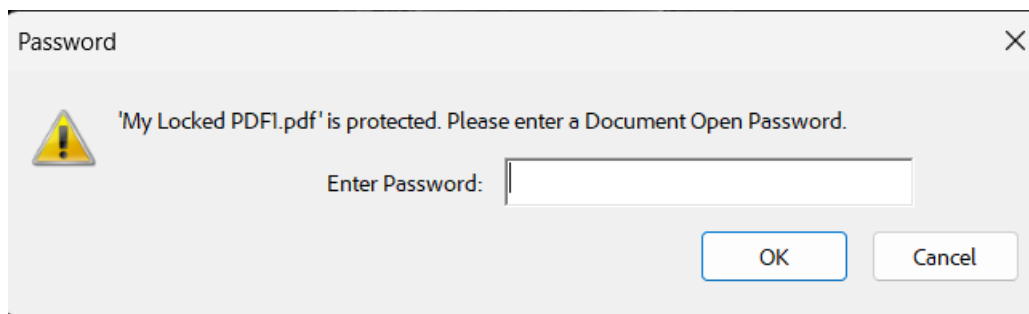
Browse ...

Notes

- Uploaded files will be deleted immediately. We do **NOT** store your files.
- This site is using pdf2john from JohnTheRipper to extract the hash
- The goal of this page is to make it very easy to convert your PDF file (.pdf) to "hashes" which hashcat or John can crack.
- Max size: 200 Mb

The previously created hash1.txt file was located and loaded into the application.

After the hash file was successfully loaded, the password-recovery operation could be initiated.

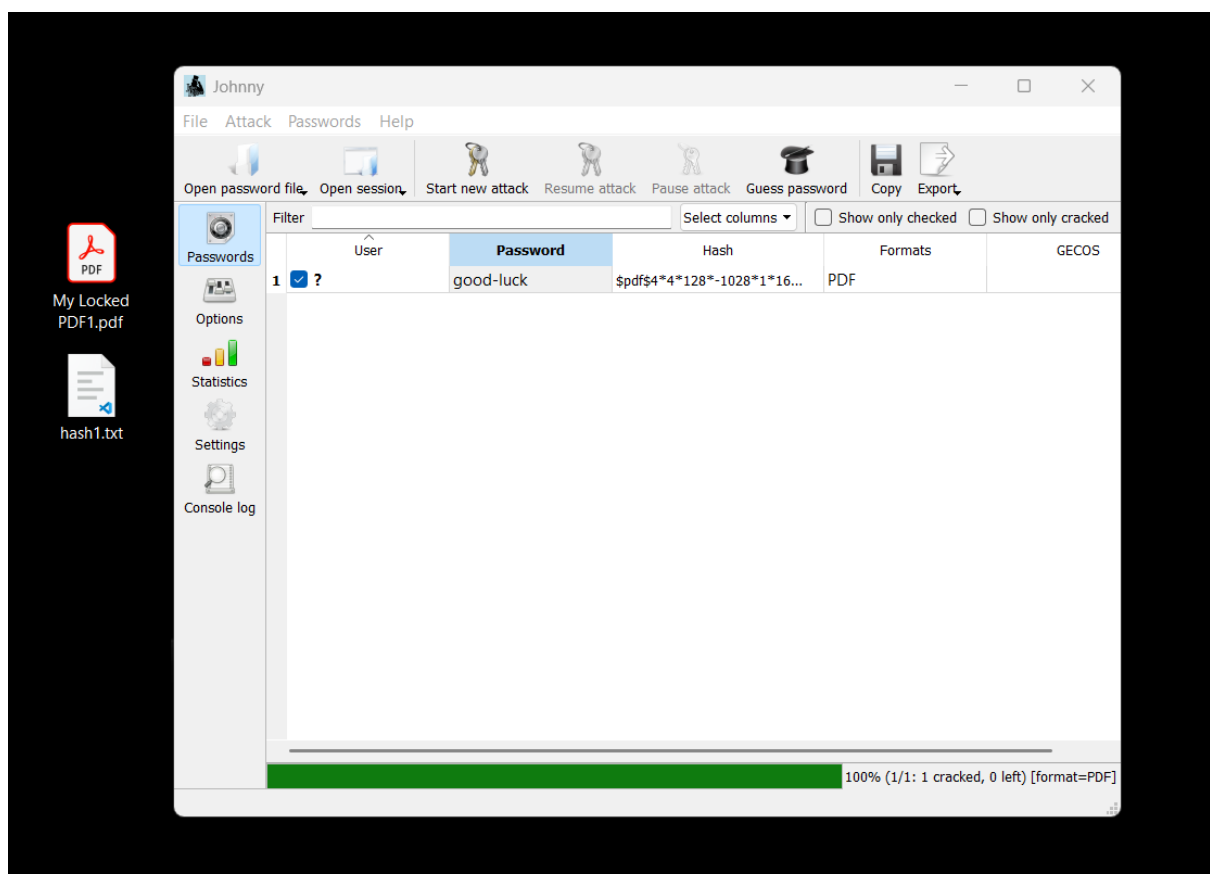


5.7 Starting the Password Recovery Process

The **Start new attack** option was selected in Johnny.

John the Ripper then processed the supplied password hash and attempted to determine the password associated with the protected PDF.

The duration of the process can vary depending on factors such as password complexity and system performance.



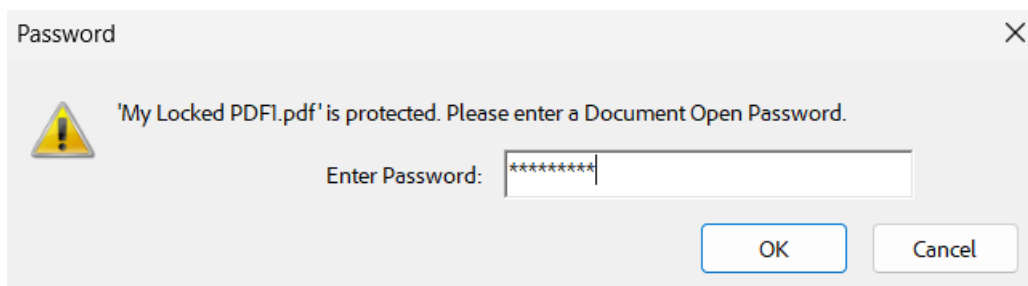
5.8 Recovered Password

The password-recovery process successfully identified the password associated with the laboratory PDF.

Recovered Password:

good-luck

The recovered value was subsequently tested against the original PDF.



5.9 Password Verification

The protected PDF was opened after completion of the JTR process.

The recovered password password1 was entered into the password prompt.

The PDF opened successfully, confirming that the recovered password was correct.



6. PROJECT MODULE 2 – PASSWORD CRACKING WITH NETWORKWALKS TOOLS

6.1 Practical Description

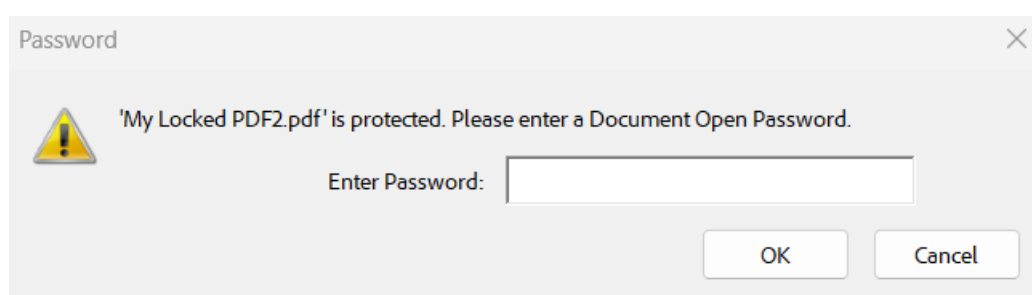
The second practical demonstrated an alternative password-recovery workflow using online tools provided by Networkwalks.

Unlike the first module, this process did not require John the Ripper or Johnny to be installed. The hash was generated through the Networkwalks Hash Calculator and then processed by the Networkwalks Password Cracker.

6.2 Obtaining the Laboratory File

The authorized password-protected PDF, My Locked PDF1.pdf, was downloaded from the Networkwalks practical lab resources.

This file was used as the input throughout the activity.



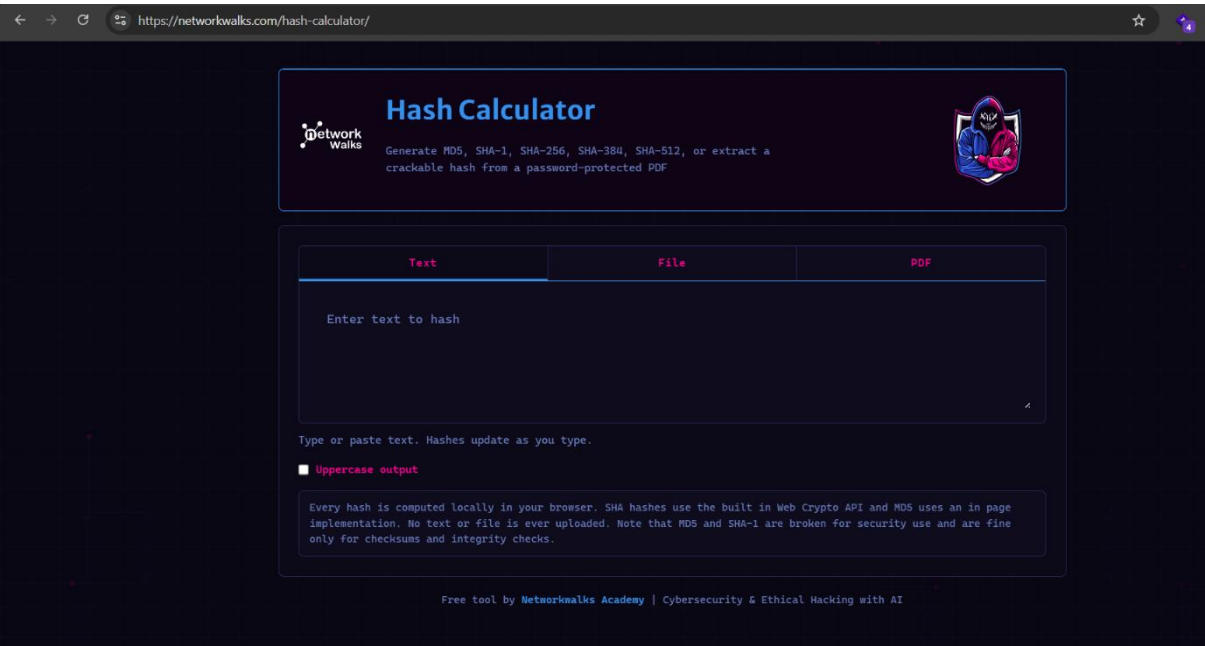
6.3 Hash Generation

The Networkwalks Hash Calculator was opened through a web browser.

The protected PDF was uploaded to the calculator.

The tool processed the file and returned a PDF password hash.

The generated value began with \$pdf\$.

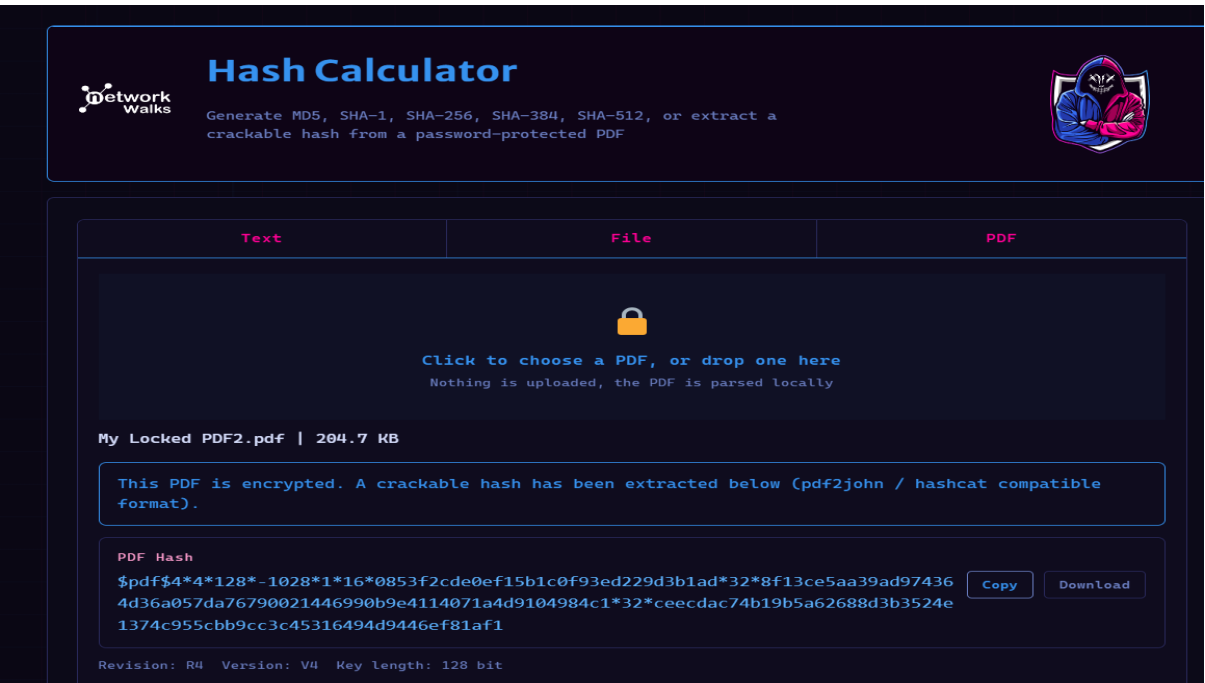


6.4 Collecting the Hash

The complete hash generated by the calculator was copied carefully.

The complete value was required because the password-cracking tool uses the supplied hash as its input.

Particular attention was given to retaining the full \$pdf\$ hash.



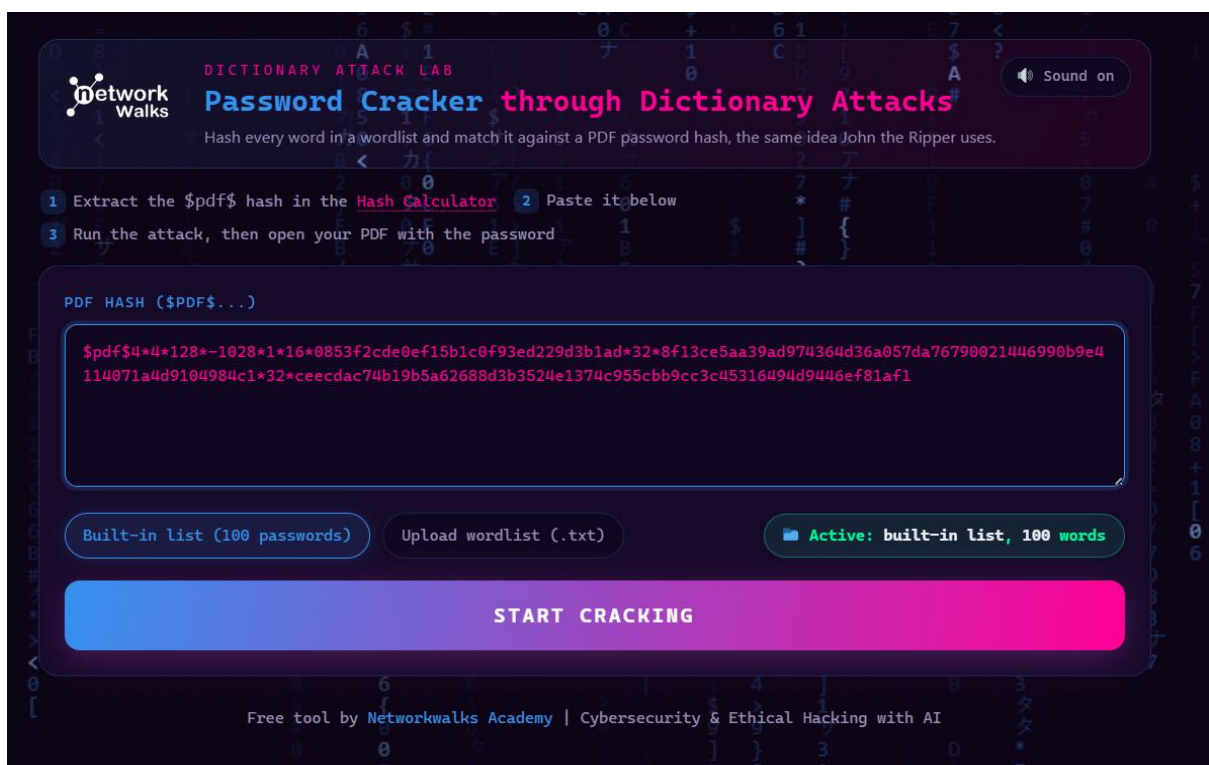
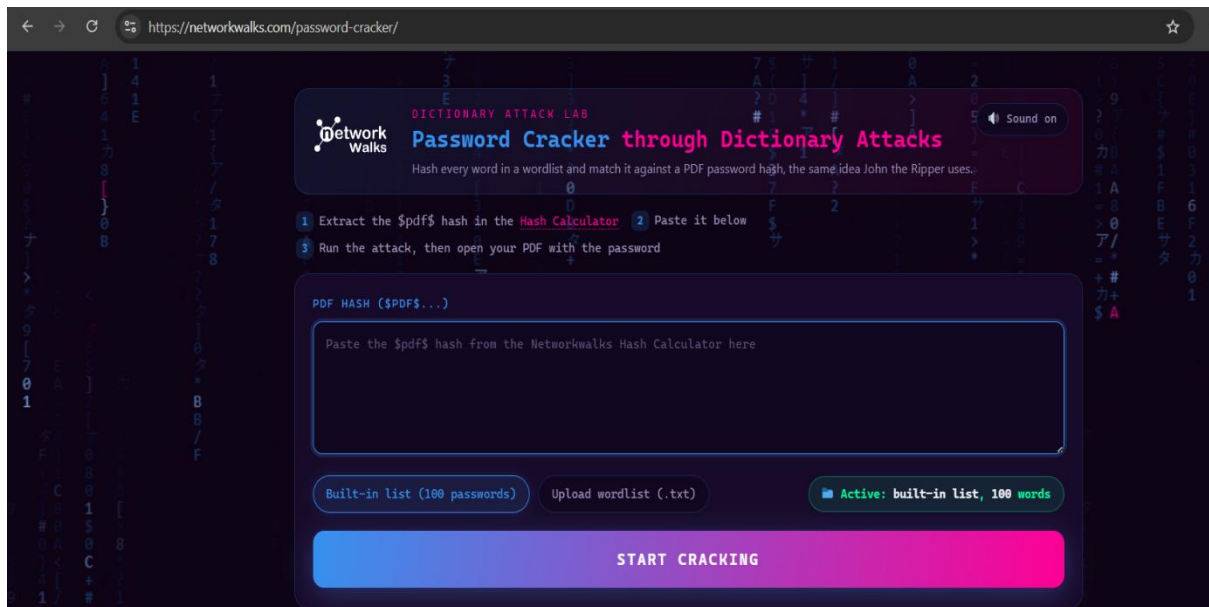
6.5 Using the Networkwalks Password Cracker

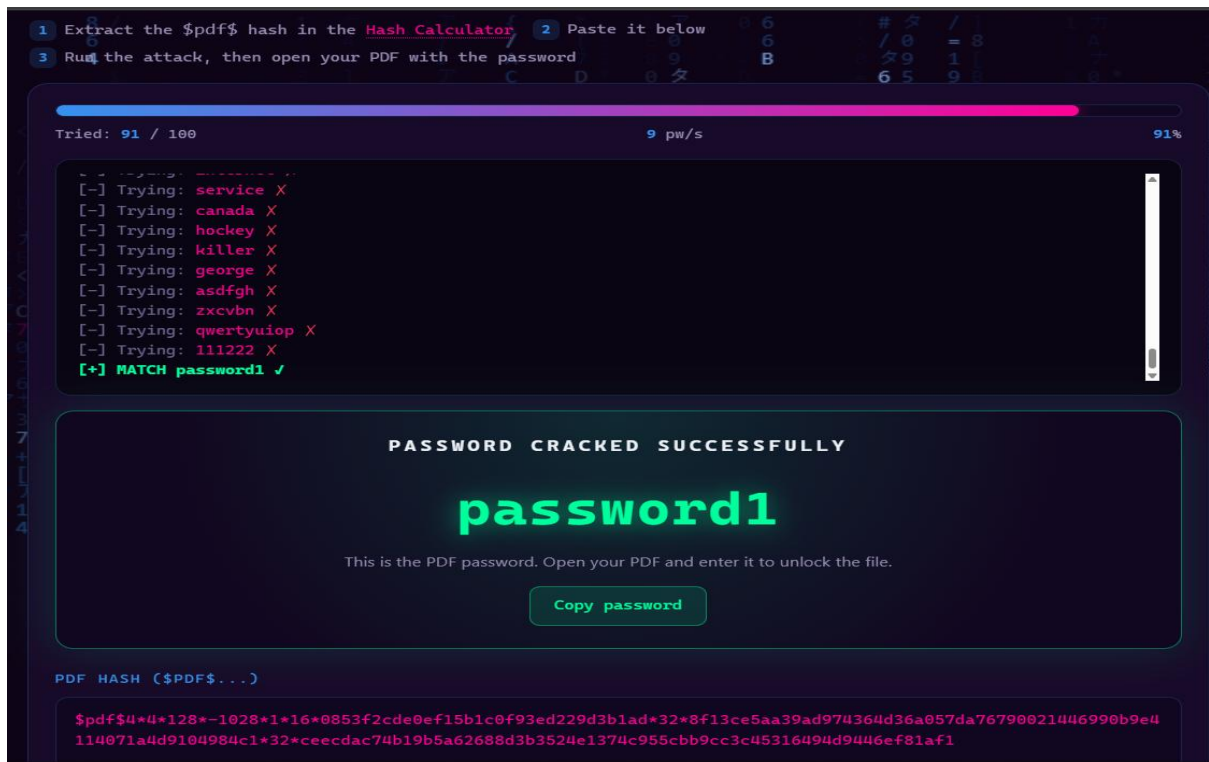
The Networkwalks Password Cracker was opened in a web browser.

The previously extracted PDF hash was entered into the appropriate field.

The password-recovery process was then initiated.

The tool attempted password candidates until a matching password was identified.





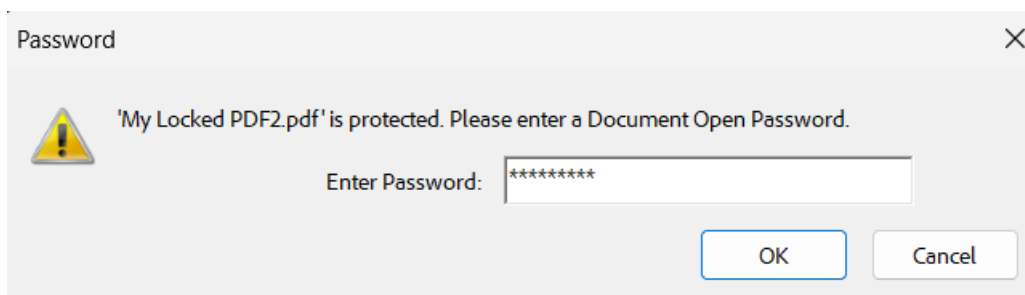
6.6 Password Recovery Result

The Password Cracker successfully recovered the password of the laboratory PDF.

Recovered Password:

password1

The recovered value was then used for verification.

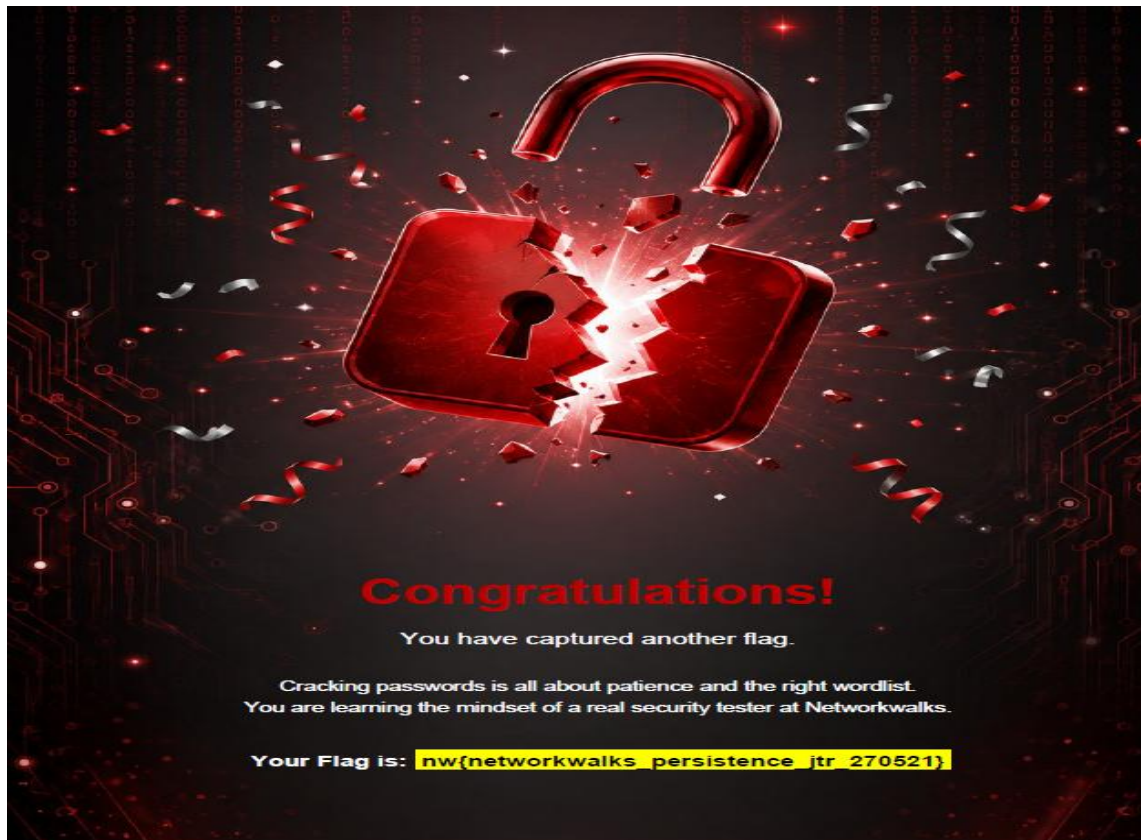


6.7 Verification of the Result

The original protected PDF was opened and the recovered password was entered.

The document opened successfully.

This confirmed that the password obtained through the Networkwalks Password Cracker matched the password protecting the PDF.



7. COMPARATIVE ANALYSIS

The two practical modules demonstrated different approaches to the same password-recovery task.

In the first module, the process involved a locally configured password-cracking application, John the Ripper, operated through Johnny.

In the second module, the password hash was generated and processed through Networkwalks' browser-based tools.

Parameter	JTR Method	Networkwalks Method
Hash Extraction	Online PDF hash extractor	Networkwalks Hash Calculator
Main Cracking Tool	John the Ripper	Networkwalks Password Cracker
User Interface	Johnny GUI	Web browser
Local Installation	Required for JTR/Johnny	Not required

Hash File	hash1.txt	Direct hash input
Target	My Locked PDF1.pdf	My Locked PDF1.pdf
Password Recovered	password1	password1
Verification	PDF opened successfully	PDF opened successfully

The comparison helped demonstrate that password-recovery workflows can be implemented using different tools while following the same general process of obtaining password-related information, processing it, and verifying the result.

8. OBSERVATIONS

The practical produced several important observations.

First, password protection does not automatically mean that a file is protected against every form of password-recovery attempt.

Second, the strength and unpredictability of the password play an important role in determining how resistant it is to password-guessing techniques.

Third, the practical showed the importance of protecting password-related information and using appropriate security controls.

Finally, the successful recovery of password1 demonstrated why simple and predictable passwords should not be used for protecting sensitive information.

9. RISK ANALYSIS AND IMPACT

The practical was performed on an authorized laboratory file and was not intended to identify a vulnerability in an external system.

However, the exercise demonstrated several security risks associated with weak passwords.

No.	Finding	Security Impact	Risk Level
1	Simple password used for file protection	May be easier to recover through password-guessing techniques	Medium
2	Predictable password pattern	Reduces the difficulty of password recovery	Medium
3	Sensitive file protected only by a weak password	Successful recovery could expose protected information	Medium

4	Password recovery possible through specialized tools	Demonstrates the need for stronger password policies	Medium
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Risk Assessment Note

These observations are based on the educational laboratory exercise.

They should not be interpreted as confirmed vulnerabilities in any external system.

No exploitation or unauthorized access was performed.

10. SECURITY RECOMMENDATIONS

Based on the practical observations, the following security measures are recommended:

10.1 Use Strong Passwords

Protected documents should use passwords that are sufficiently long and difficult to predict.

10.2 Avoid Common Passwords

Passwords such as password1, dictionary words, names, or simple numerical combinations should be avoided.

10.3 Use Unique Passwords

The same password should not be reused across multiple documents, accounts, or services.

10.4 Use Password Managers

Password managers can help generate and securely store strong and unique passwords.

10.5 Protect Sensitive Documents

Confidential files should be protected using appropriate encryption and access-control mechanisms.

10.6 Regularly Review Password Policies

Organizations should periodically review their password policies and ensure that weak password practices are discouraged.

10.7 Conduct Authorized Security Testing

Password-strength assessments should be performed regularly, but only against systems and files for which proper authorization has been obtained.

11. SKILLS DEVELOPED

During these practical activities, I developed practical knowledge in the following areas:

- Password security
- Password hash concepts
- PDF password protection
- John the Ripper
- Johnny GUI
- Hash extraction
- Password recovery
- Networkwalks Hash Calculator
- Networkwalks Password Cracker
- Security testing methodology
- Evidence collection
- Technical documentation
- Ethical cybersecurity practices

12. EVIDENCE DOCUMENTATION

The following screenshots were collected during the practical exercises.

Module 1 – JTR

1. John the Ripper download/setup
2. Johnny installation
3. Johnny settings
4. john.exe configuration
5. PDF hash extraction
6. Extracted PDF hash
7. hash1.txt
8. Password file loaded in Johnny
9. JTR attack started
10. Password recovered
11. PDF successfully unlocked

Module 2 – Networkwalks Tools

1. Downloaded My Locked PDF1.pdf
2. Networkwalks Hash Calculator
3. Uploaded PDF
4. Generated \$pdf\$ hash
5. Complete hash
6. Networkwalks Password Cracker
7. Password-cracking process
8. Recovered password
9. Password entered into PDF
10. Successfully opened PDF

13. ETHICAL CONSIDERATIONS

Password-cracking technology has legitimate applications in cybersecurity, including password auditing, security research, incident investigation, and educational training.

At the same time, these techniques can be misused if they are applied without authorization.

For this reason, the activities documented in this report were limited to the PDF file provided for the Networkwalks practical.

The exercise was performed for educational purposes, and no attempt was made to access unauthorized files, accounts, systems, or services.

14. OVERALL LEARNING OUTCOME

The Week 3 practical provided hands-on experience with the process of password recovery and password-strength assessment.

The first module introduced me to John the Ripper and Johnnny, giving me practical experience with configuring a password-cracking environment, preparing a PDF hash, creating a hash file, and starting a password-recovery operation.

The second module provided an alternative browser-based approach using Networkwalks' Hash Calculator and Password Cracker.

Working with both approaches helped me understand that the password-recovery process involves multiple stages rather than simply entering a password into a tool. The hash must first be obtained in the correct format before it can be processed by the appropriate cracking tool.

The practical also highlighted the importance of strong passwords. The successful recovery of the password password1 demonstrated how predictable credentials can provide inadequate protection for sensitive files.

15. CONCLUSION

During Week 3 of my Cybersecurity Internship Program, I completed two practical exercises focused on password cracking and password security.

In Project Module 1, I used John the Ripper and Johnny to process the password information extracted from the authorized PDF file. The hash was stored in hash1.txt, loaded into Johnny, and used for the password-recovery process.

In Project Module 2, I performed the same general password-recovery workflow using the Networkwalks Hash Calculator and Networkwalks Password Cracker. The PDF hash was generated online and subsequently supplied to the password-cracking tool.

Both methods successfully recovered the password **password1**, and the result was verified by opening the protected PDF.

Overall, these exercises improved my practical understanding of password hashes, password-recovery tools, password strength, and security testing methodology. The activities also reinforced the importance of using strong, unique, and unpredictable passwords and conducting cybersecurity testing only within an authorized scope.

16. AUTHOR

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Week: 03

Project: Password Cracking with JTR & Networkwalks Tools